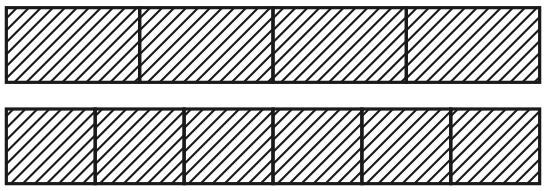
Fractions and Wholes

All the pieces make 1 whole. More pieces go past 1.

All the pieces: 1 whole

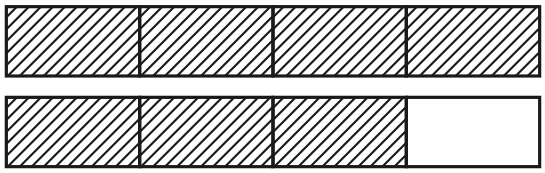


$\frac{4}{4} = 1$

$\frac{6}{6} = 1$

Top and bottom the same? Then it's 1 whole.

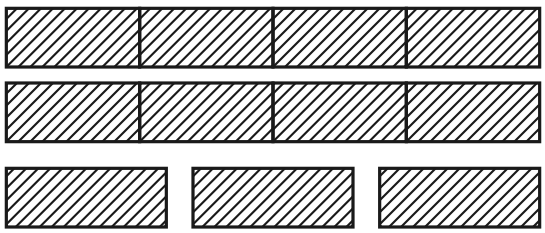
More than 1 whole



$\frac{7}{4} = 1 \text{ whole and } \frac{3}{4}$

Count on past 4/4: 5/4, 6/4, 7/4.

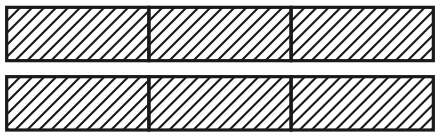
Whole numbers are fractions too



$\frac{8}{4} = 2$ 8 fourths fill 2 bars.

$\frac{3}{1} = 3$ 3 pieces, each a whole.

Your turn!



$\frac{6}{3} = 2$

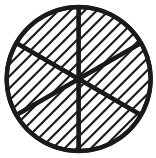
For teachers

Goal: b/b = 1, fractions bigger than 1, and whole numbers as fractions (3.NF.3c). Say "How many pieces make one whole bar? Count on past it." No mixed-number rules yet: 7/4 is 1 whole and 3/4.

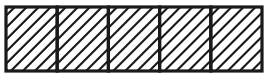
Fractions and Wholes: Practice

When the top number matches the bottom, that's 1 whole.

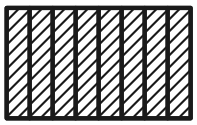
Part 1 One whole Every part is shaded. How many pieces?

1) 

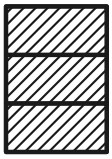
$\frac{6}{6} = 1$

2) 

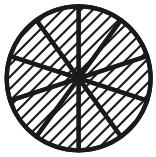
$\frac{5}{5} = 1$

3) 

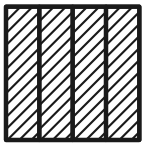
$\frac{8}{8} = 1$

4) 

$\frac{3}{3} = 1$

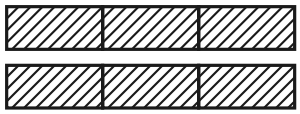
5) 

$\frac{10}{10} = 1$

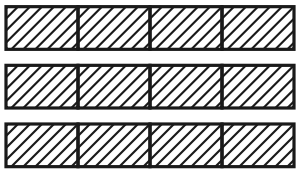
6) 

$\frac{4}{4} = 1$

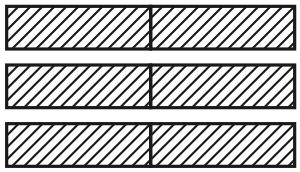
Part 2 How many wholes? Count the pieces, then the whole bars.

7) 

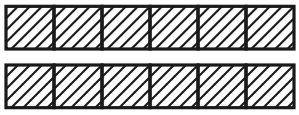
$\frac{6}{3} = 2$

8) 

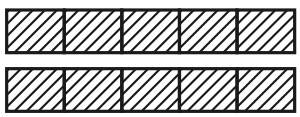
$\frac{12}{4} = 3$

9) 

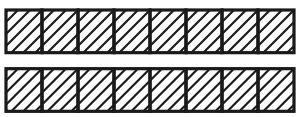
$\frac{6}{2} = 3$

10) 

$\frac{12}{6} = 2$

11) 

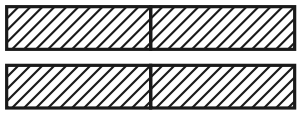
$\frac{10}{5} = 2$

12) 

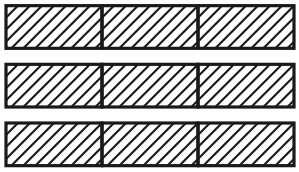
$\frac{16}{8} = 2$

Fractions and Wholes: Practice (continued)

Part 2 How many wholes? (continued)

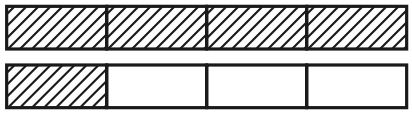
13) 

$\frac{4}{2} = 2$

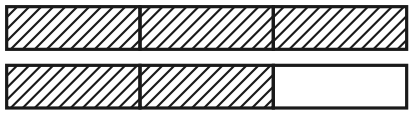
14) 

$\frac{9}{3} = 3$

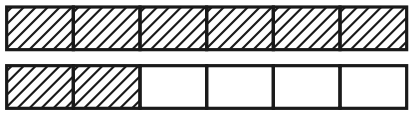
Part 3 More than one whole Full bars first, then the pieces left.

15) 

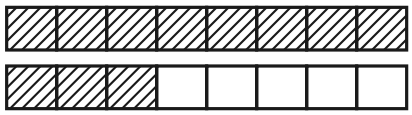
$\frac{5}{4} = 1 \text{ whole and } \frac{1}{4}$

16) 

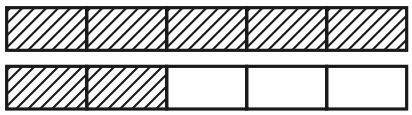
$\frac{5}{3} = 1 \text{ whole and } \frac{2}{3}$

17) 

$\frac{8}{6} = 1 \text{ whole and } \frac{2}{6}$

18) 

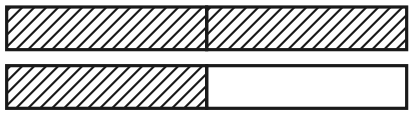
$\frac{11}{8} = 1 \text{ whole and } \frac{3}{8}$

19) 

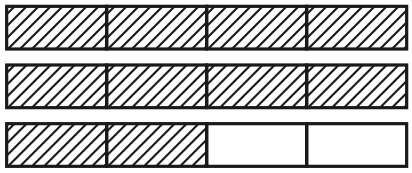
$\frac{7}{5} = 1 \text{ whole and } \frac{2}{5}$

Fractions and Wholes: Practice (continued)

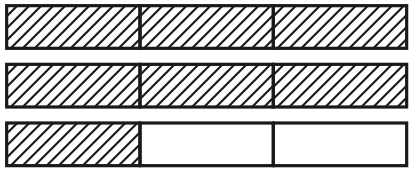
Part 3 More than one whole (continued)

20) 

$\frac{3}{2}$ = 1 whole and $\frac{1}{2}$

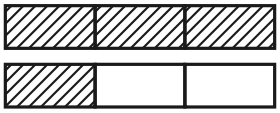
21) 

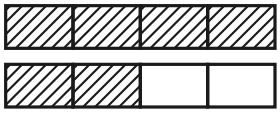
$\frac{10}{4}$ = 2 wholes and $\frac{2}{4}$

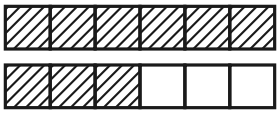
22) 

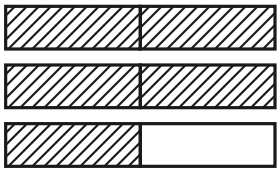
$\frac{7}{3}$ = 2 wholes and $\frac{1}{3}$

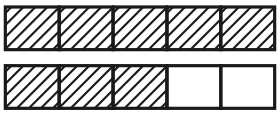
Part 4 Shade it

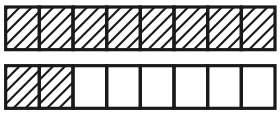
23) Shade $\frac{4}{3}$ 

24) Shade $\frac{6}{4}$ 

25) Shade $\frac{9}{6}$ 

26) Shade $\frac{5}{2}$ 

27) Shade $\frac{8}{5}$ 

28) Shade $\frac{10}{8}$ 

Fractions and Wholes: Practice (continued)

Part 5 Less than, equal to, or more than 1? Circle one.

29) $\frac{5}{4}$ less than 1 = 1 **(more than 1)**

30) $\frac{3}{8}$ **(less than 1)** = 1 more than 1

31) $\frac{6}{6}$ less than 1 **(= 1)** more than 1

32) $\frac{2}{3}$ **(less than 1)** = 1 more than 1

33) $\frac{9}{8}$ less than 1 = 1 **(more than 1)**

34) $\frac{10}{10}$ less than 1 **(= 1)** more than 1

35) $\frac{7}{5}$ less than 1 = 1 **(more than 1)**

36) $\frac{4}{6}$ **(less than 1)** = 1 more than 1

37) $\frac{3}{3}$ less than 1 **(= 1)** more than 1

38) $\frac{12}{10}$ less than 1 = 1 **(more than 1)**

Part 6 One whole No help this time.

39) $\frac{5}{5} = \boxed{1}$

40) $\frac{\boxed{8}}{8} = 1$

41) $\frac{10}{10} = \boxed{1}$

42) $\frac{\boxed{3}}{3} = 1$

43) $\frac{2}{2} = \boxed{1}$

44) $\frac{\boxed{6}}{6} = 1$

Part 7 Whole numbers No help this time.

45) $\frac{12}{4} = \boxed{3}$

46) $\frac{10}{5} = \boxed{2}$

47) $\frac{6}{2} = \boxed{3}$

48) $\frac{9}{3} = \boxed{3}$

49) $\frac{16}{8} = \boxed{2}$

50) $\frac{12}{6} = \boxed{2}$

51) $\frac{8}{2} = \boxed{4}$

52) $\frac{15}{5} = \boxed{3}$

53) $\frac{20}{10} = \boxed{2}$

Fractions and Wholes: Practice (continued)

Part 7 Whole numbers (continued)

54) $\frac{4}{1} = \boxed{4}$

55) $\frac{18}{6} = \boxed{3}$

56) $\frac{6}{3} = \boxed{2}$

Part 8 Write the whole number as a fraction No help this time.

57) $\frac{\boxed{8}}{2} = 4$

58) $\frac{\boxed{12}}{3} = 4$

59) $\frac{\boxed{16}}{4} = 4$

60) $\frac{\boxed{10}}{2} = 5$

61) $\frac{\boxed{24}}{8} = 3$

62) $\frac{\boxed{30}}{10} = 3$

63) $\frac{\boxed{6}}{1} = 6$

64) $\frac{\boxed{20}}{5} = 4$

65) $\frac{\boxed{24}}{6} = 4$

Part 9 More than one whole No help this time.

66) $\frac{9}{8}$ = 1 whole and $\frac{\boxed{1}}{8}$

67) $\frac{7}{5}$ = 1 whole and $\frac{\boxed{2}}{5}$

68) $\frac{11}{4}$ = 2 wholes and $\frac{\boxed{3}}{4}$

69) $\frac{8}{6}$ = 1 whole and $\frac{\boxed{2}}{6}$

70) $\frac{13}{10}$ = 1 whole and $\frac{\boxed{3}}{10}$

71) $\frac{5}{2}$ = 2 wholes and $\frac{\boxed{1}}{2}$

72) $\frac{10}{3}$ = 3 wholes and $\frac{\boxed{1}}{3}$

73) $\frac{11}{6}$ = 1 whole and $\frac{\boxed{5}}{6}$

74) $\frac{17}{8}$ = 2 wholes and $\frac{\boxed{1}}{8}$

75) $\frac{9}{4}$ = 2 wholes and $\frac{\boxed{1}}{4}$

Fractions and Wholes: Practice (continued)

Part 10 Count past 1 No help this time.

76) $\frac{3}{4}\ \frac{4}{4}\ \boxed{\frac{5}{4}}\ \boxed{\frac{6}{4}}\ \frac{7}{4}$

77) $\frac{4}{5}\ \frac{5}{5}\ \boxed{\frac{6}{5}}\ \boxed{\frac{7}{5}}$

78) $\frac{2}{3}\ \frac{3}{3}\ \frac{4}{3}\ \boxed{\frac{5}{3}}\ \boxed{\frac{6}{3}}$

79) $\frac{7}{8}\ \frac{8}{8}\ \boxed{\frac{9}{8}}\ \boxed{\frac{10}{8}}$

80) $\frac{1}{2}\ \frac{2}{2}\ \boxed{\frac{3}{2}}\ \boxed{\frac{4}{2}}\ \frac{5}{2}$

81) $\frac{5}{6}\ \frac{6}{6}\ \frac{7}{6}\ \boxed{\frac{8}{6}}\ \boxed{\frac{9}{6}}$

Part 11 Remember? Problems from earlier packets.

82) 9 cm by 3 cm: area = [27] sq cm

83) Round 436 to the nearest ten: [440]

84) two tenths = $\boxed{\frac{2}{10}}$

85) 6 groups of 8 = [48]

86) 4 pieces of $\frac{1}{8}$ = $\boxed{\frac{4}{8}}$

87) Round 819 to the nearest hundred: [800]

Part 12 Word problems Draw the bars.

88) Ana has 2 pizzas. Each one is cut into 4 equal slices. How many fourths are there in all?

Number sentence 2 wholes = 8/4

Answer [8] fourths

Draw or show your work

89) Lee needs 1 cup of milk but only has a 1/3 cup scoop. How many scoops make 1 cup?

Number sentence 3/3 = 1

Answer [3] scoops

Draw or show your work

Fractions and Wholes: Practice (continued)

Part 12 Word problems (continued)

90) Max eats 6 halves of apples. How many whole apples is that?

Number sentence 6/2 = 3

Answer 3 apples

Draw or show your work

91) Sam has 7 pieces of two same-size bars. Each bar has 4 pieces. Is that more than 1 bar?

Number sentence 7/4 = 1 whole and 3/4

Answer yes

Draw or show your work

92) Challenge: Jo has 10 eighths of a pie. How many whole pies is that, and how many eighths?

Number sentence 10/8 = 1 whole and 2/8

Answer 1 and 2/8 pies

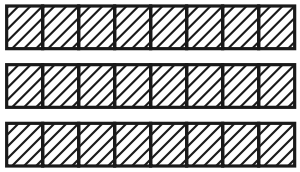
Draw or show your work

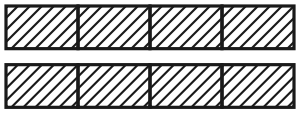
Check-In: Fractions and Wholes

Show what you know. No help this time.

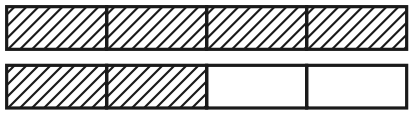
Mastery: 8 of 10 (80%)

Part 1 How many wholes?

1)  $\frac{24}{8} = 3$

2)  $\frac{8}{4} = 2$

Part 2 More than one whole

3)  $\frac{6}{4}$ = 1 whole and $\frac{2}{4}$

Part 3 Less than, equal to, or more than 1?

4) $\frac{5}{3}$ less than 1 = 1 **more than 1**

5) $\frac{11}{6}$ less than 1 = 1 **more than 1**

Part 4 Wholes

6) $\frac{8}{8} = 1$

7) $\frac{18}{6} = 3$

8) $\frac{30}{10} = 3$

Part 5 Past one whole

9) $\frac{15}{8}$ = 1 whole and $\frac{7}{8}$

10) $\frac{8}{3}$ = 2 wholes and $\frac{2}{3}$